

SafeOT-Dual: one price rule, two views

Penalty view

$$\max_{\pi} J_r(\pi) - \lambda^T J_c(\pi)$$

Exact settings

$$c = 0$$

Fixed β

fixed penalty
 $\lambda = \beta$

Adaptive β

Lagrangian feedback

Joint constraints

$$C_1, b_1$$

$$C_2, b_2$$

...

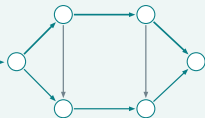
$$C_K, b_K$$

Hard support H



$$F_H = 0$$

One graph problem



$$\min_F - \langle A, F \rangle + \varepsilon KL(F \| \hat{F})$$

flow balance

$$\langle C_k, F \rangle \leq b_k$$

Shared safety price

$$\lambda^*$$

Clip + filter

$$\lambda_k = \lambda_k^* + \beta_k$$

graph price

feedback

$$\lambda_1$$

$$\lambda_2$$

...

$$\lambda_K$$

Constraint view

$$\max_{\pi} J_r(\pi) \text{ s.t. } J_c(\pi) \leq B$$

Graph-price limits under assumptions

$$\varepsilon \rightarrow 0$$

LP-derived price

$$\varepsilon \rightarrow \infty$$

clipped active-set price

One policy-update rule

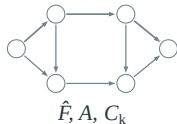
$$\tilde{A} = (A_r - \sum_k \lambda_k A_{c_k}) / (1 + \sum_k \lambda_k)$$



PPO / TRPO

Graph-space price construction

Empirical graph



Joint budget solve

 C_1, b_1 C_2, b_2 C_K, b_K $b_k = B_k/L$ $F_H = 0$

Entropic flow problem

$$\min_F - \langle A, F \rangle + \varepsilon KL(F \parallel \hat{F})$$

$$\text{flow balance} \quad \langle C_k, F \rangle \leq b_k$$

Dual solver

Newton in u Bisection in λ λ^*

Clip + filter

$$\lambda^- \leftarrow (1-\alpha)\lambda^- + \alpha \text{clip}(\lambda^*, 0, c)$$



transitions + costs

Only prices enter the policy update

Policy-space learning

Policy rollout

Episode $\bar{J}$

Cost feedback

$$\beta \leftarrow [\beta + \eta(\bar{J} - B)]_+$$

 β λ^- λ

One priced update

$$\frac{A_r - \sum_k \lambda_k A_{c_k}}{1 + \sum_k \lambda_k}$$

PPO / TRPO

GAE: A_r, A_c

next rollout

 π_{t+1}